When Is the Mean Nearly Normal?

AP Statistics · Topic 5.3 (CED 2.12) · B: 2026-11-13 · E: 2026-12-09 · TEACHER KEY

CED TETHER

- **Skill 3.C** — Describe probability distributions.
- **Enduring Understanding UNC-3** — Probabilistic reasoning allows us to anticipate patterns in data.
- **LO UNC-3.H** — Estimate sampling distributions using simulation. [Skill 3.C]
- **EK UNC-3.H.1** — A sampling distribution of a statistic is the distribution of values for the statistic for all possible samples of a given size from a given population.
- **EK UNC-3.H.2** — The central limit theorem (CLT) states that when the sample size is sufficiently large, a sampling distribution of the mean of a random variable will be approximately normally distributed.
- **EK UNC-3.H.3** — The central limit theorem requires that the sample values are independent of each other and that n is sufficiently large.
- **EK UNC-3.H.4** — A randomization distribution is a collection of statistics generated by simulation assuming known values for the parameters.
- **EK UNC-3.H.5** — The sampling distribution of a statistic can be simulated by generating repeated random samples from a population.

Essential question: How do population shape and sample size determine whether a normal model is trustworthy for sample means?

DOK-3 skill: 4.B · **FRQ pattern:** `clt-when-does-normality-apply`

DOK rationale: Part (c) coordinates population shape, sample size, independence, and sampling-distribution parameters to adjudicate two normality claims and bound which probability model a reader may use.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK:** 1 → 3 **Minutes:** 45

Teacher does

- Board slide up at the bell. Ask for a sample-size choice before displaying a rule of thumb.
- Begin the 5.3 video follow-along at minute 5.
- Return to the board slide at minute 33. Circulate and require students to name which distribution becomes approximately normal.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Choose n equals 4 or n equals 40 and cite population shape or sample size.
- Complete the follow-along about simulated sampling distributions, the CLT, and randomization distributions.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- Does averaging four values erase a long right tail automatically?

- Which distribution does the CLT describe: commuters or sample means?
- What happens to $15/\sqrt{n}$ when n grows from 4 to 40?
- Which probability calculation becomes questionable if the n equals 4 shape is not normal?

Adult role
Solo teacher. Circulate 5 pairs. Require shape, sample size, and independence in every normality defense; redirect claims that the population itself changes shape.

[WARN] WATCH FOR

- Invoking the CLT for n equals 4 without considering the strong population skew.
- Saying forty individual commute times have a normal distribution.
- Using 15 minutes, rather than $15/\sqrt{n}$, for the spread of sample means.

SCORING PART (c) — E / P / I

Expected elements

- **rejects-small-n-normality** (required) — Rejects the automatic normal claim for n equals 4 because the source population is strongly right-skewed
- **supports-large-n-normality** (required) — Uses independence and sufficiently large n equals 40 to invoke the central limit theorem
- **uses-large-n-spread** (required) — States that the n equals 40 sampling distribution is centered at 25 minutes with standard deviation about 2.37 minutes
- **distinguishes-distributions** (required) — Clarifies that the CLT concerns the distribution of sample means, not the distribution of individual commute times
- **states-model-consequence** (required) — Explains that a normal approximation at n equals 4 could give inaccurate tail probabilities

E	Rejects normality at n equals 4, supports it at n equals 40 with independence and the CLT, uses the 2.37-minute standard deviation, distinguishes the two distributions, and explains the tail-probability consequence.
P	Chooses the appropriate sample sizes and mentions skew and the CLT but omits independence, the computed spread, the distribution distinction, or the consequence for probability calculations.
I	Says every average is normal, says the population itself becomes normal at n equals 40, or uses 15 minutes as the sampling-distribution standard deviation for both sample sizes.

Common mistakes

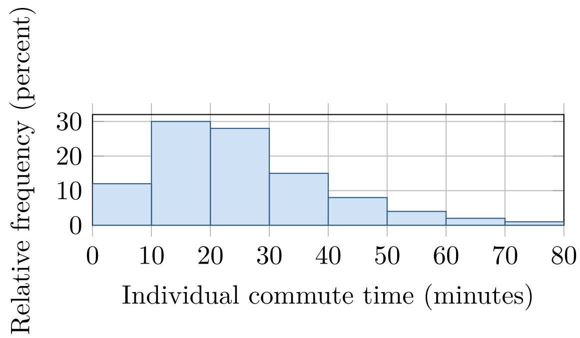
- Treating the central limit theorem as applying at every sample size
- Saying the individual commute-time population becomes normal when n increases
- Dividing the mean, rather than the standard deviation, by $\sqrt{n}$
- Using the number of repeated samples instead of the observations per sample as n

[STAR] TODAY'S PROBLEM — annotated

Individual commute times in a very large city are strongly right-skewed, with population mean $\mu = 25$ minutes and standard deviation $\sigma = 15$ minutes. Consider independent random samples.

Rae: "Any average is approximately normal, even for $n = 4$."

Sol: "With this long right tail, $n = 4$ may be too small; $n = 40$ is more defensible."



FIRST TAKE (5 min)

For which sample size, $n = 4$ or $n = 40$, would you trust a normal model for the sample mean? Name one feature behind your choice.
Key: Not graded. Expect 'averages are normal' as an initial rule; the revision should add population shape and sufficiently large n.

Rules callout "CLT: SHAPE, SIZE, AND THE RIGHT DISTRIBUTION (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** Identify the population shape and state the population mean and standard deviation with units.

Key: The population distribution of individual commute times is strongly right-skewed. Its mean is $\mu = 25$ minutes and its standard deviation is $\sigma = 15$ minutes.

DOK 2 **(b)** For $n = 4$ and $n = 40$, calculate the mean and standard deviation of the sampling distribution of $\bar{x}$.

Key: For either sample size, $\mu_{\bar{x}} = \mu = 25$ minutes. For $n = 4$, $\sigma_{\bar{x}} = 15/\sqrt{4} = 7.5$ minutes. For $n = 40$, $\sigma_{\bar{x}} = 15/\sqrt{40} \approx 2.3717$ minutes, or about 2.37 minutes.

DOK 3 **(c)** Adjudicate both claims for $n = 4$ and $n = 40$. Cite the histogram shape, independence, and the CLT's sample-size requirement; use the $n = 40$ standard deviation; and explain one consequence of ignoring skew when using a normal model at $n = 4$.

Key: Rae's claim is not warranted at $n = 4$: averaging only four observations from a strongly right-skewed population need not remove the skew. Sol is warranted at $n = 40$: independence and the large sample size let the CLT support an approximately normal sampling distribution centered at 25 minutes with standard deviation $15/\sqrt{40} \approx 2.37$ minutes. The CLT concerns sample means, not individual commute times. Ignoring the population skew at $n = 4$ could make normal-curve tail probabilities inaccurate, making rare sample means look too common or too rare.

EXIT

One thing the video changed about my first take: ______________________________